



DISTRIBUTED SOVEREIGN AI
INFRASTRUCTURE NODES TECHNOLOGY

DEMONSTRATOR NODE 01 BLUEPRINT

BEDBURG-HAU,
GERMANY

EMBEDDED SITE
MASTERPLAN



AI
INFRASTRUCTURE



RENEWABLE
ENERGY



BTES
SYSTEM



HEAT
RECOVERY



DIGITAL
SOVEREIGNTY



CYBER
SECURITY



EUROPEAN
CONNECTIVITY

BUILDING GERMANY'S SOVEREIGN AI INFRASTRUCTURE BACKBONE



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DSAIN

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DEMONSTRATOR NODE 01

MASTERPLAN

Bedburg-Hau, North Rhine-Westphalia, Germany

Updated Strategic & Technical Edition | Version 3.0 | August 2026

A physical reference architecture for sovereign AI infrastructure — developed in stages, validated through technology prototypes, and designed as the future operational blueprint for the DSAINT federation.

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Based on the uploaded DSAINT Demonstrator Node 01 Masterplan Version 2.0, with the current DSAINT development strategy incorporated.

1. Document Basis and Strategic Update

This Masterplan is based on the uploaded DSAINT Demonstrator Node 01 Masterplan Version 2.0. The source identifies Bedburg-Hau as the location, defines Node 01 as a sovereign AI infrastructure demonstrator and national blueprint, specifies a 2 MW initial IT capacity expandable to 6 MW, Tier III infrastructure, N+1 redundancy, direct-to-chip liquid cooling and a target PUE of ≤ 1.20 , and describes the site's operational zones and federation functions.

The updated edition retains those source-derived specifications and functions while incorporating the current DSAINT strategic direction: Node 01 should be developed as a staged physical demonstrator whose final architecture is informed by evidence from the DSAINT workload-orchestration and federation prototype. Large-scale construction is therefore preceded by technical, operational, economic and regulatory validation.

Node 01 remains the physical heart of the long-term DSAINT vision — but it is no longer assumed to be the first proof that DSAINT's technology works.

2. Project Identification

Item	Masterplan definition
Project	DSAINTE Demonstrator Node 01
Program	DSAINTE Sovereign AI Infrastructure Ecosystem
Location	Bedburg-Hau, North Rhine-Westphalia, Germany
Role	National Demonstrator and Operational Blueprint
Initial IT capacity	2 MW
Expansion capacity	6 MW
Target infrastructure standard	Tier III
Cooling	Direct-to-Chip Liquid Cooling
Redundancy	N+1
Target PUE	≤ 1.20
Connectivity	Multi-carrier fiber
Security model	Security by Design
Operational role	National federation coordination and sovereign AI infrastructure demonstrator
Planning commissioning target	2027 — subject to validation, permitting, financing and implementation gates

3. Executive Summary

Demonstrator Node 01 is conceived as the first physical deployment of DSAINT's sovereign AI infrastructure architecture in Bedburg-Hau, Germany. The uploaded source positions the facility as a Sovereign AI Infrastructure Demonstrator, Community Compute Federation Coordination Center, Reverse Co-Hosting Operations Hub, national blueprint and foundation for a German sovereign AI federation.

In the updated masterplan, Node 01 has an additional role: it is the physical validation environment for the DSAINT federation and AI workload orchestration architecture. The facility will provide the high-density compute, network, power, cooling, security and operational environment required to test the transition from software-defined federation to production-grade infrastructure.

The development principle is staged: prove the orchestration architecture in controlled environments; validate operating assumptions with partners; complete site and infrastructure feasibility; then construct and commission the physical demonstrator.

Objective: build the reference node only after the technical and organizational assumptions required to operate it have been sufficiently validated.

4. Strategic Purpose

Purpose	Node 01 contribution
Sovereign AI Infrastructure	Secure, resilient and environmentally responsible AI-ready infrastructure.
Federation Coordination	Coordinate trusted DSAINT nodes and future Community Compute Federation participants.
Workload Orchestration	Provide the physical environment for distributed AI workload scheduling and execution.
Contribution Accounting	Measure and validate future participating-node contributions.
Innovation Platform	Support municipalities, enterprises, universities and research institutions.
National Blueprint	Establish repeatable technical and operational standards for future DSAINT Core Nodes.
Regional Economic Platform	Support skilled employment, research collaboration and municipal innovation.

5. Design Principles

- Sovereignty: infrastructure and operations designed for European/German strategic independence.
- Resilience: distributed architecture, N+1 redundancy and multi-carrier connectivity.
- Security by design: identity, access, network and workload security integrated from the beginning.
- Energy responsibility: target PUE ≤ 1.20 , liquid cooling and heat recovery opportunities.
- Modularity: physical systems should support staged expansion from 2 MW to 6 MW.
- Observability: power, cooling, compute, network and workload performance must be measurable.
- Interoperability: standards-based interfaces should allow federation with research, enterprise and future community nodes.
- Operational readiness: NOC/SOC, maintenance, logistics and security are treated as first-class infrastructure.
- Evidence-based development: major technical decisions should be supported by prototype results and feasibility studies.

6. Technical Architecture

6.1 Infrastructure Core

- AI-ready infrastructure halls
- Server and accelerator environments

- Network systems and high-speed interconnects
- Power distribution systems
- Monitoring and telemetry systems
- Secure management infrastructure

6.2 DSAINT Orchestration and Federation Plane

The physical Node 01 should host and validate the DSAINT control-plane functions developed through the software federation prototype.

Capability	Function
Node identity	Authenticate infrastructure participants and establish trust.
Capability discovery	Inventory CPU, GPU, memory, storage, network and policy capabilities.
Workload submission	Receive AI workloads and associated requirements.
Scheduling	Select appropriate compute resources based on capacity, policy, performance and other constraints.
Monitoring	Observe node health, workload status, utilization and performance.
Failure handling	Detect node failure and support workload recovery or rescheduling.
Accounting	Measure future federation contributions and service consumption.
Security policy	Apply workload and node trust policies.

7. Capacity and Infrastructure Specifications

Category	Specification / target
Initial IT capacity	2 MW
Future expansion	Up to 6 MW
Infrastructure standard	Tier III
Redundancy	N+1
Cooling	Direct-to-Chip Liquid Cooling
Target PUE	≤ 1.20
Connectivity	Multi-carrier fiber
Security	Security by Design / Zero Trust-oriented architecture
Operational model	24/7 infrastructure and federation operations

The numerical specifications above are derived from the uploaded Masterplan. Detailed engineering values for rack density, electrical topology, UPS autonomy, generator capacity, cooling water temperatures, storage capacity and network bandwidth are not defined in the source and therefore remain engineering-design work packages.

8. Masterplan Site Zones

Zone	Function	Key contents from source
A – AI Infrastructure Core	Primary AI infrastructure hall	AI-ready infrastructure, server environments, network systems, power distribution, monitoring
B – Operations & Federation Center	Administrative and operational	NOC, SOC, federation management,

	headquarters	Citizen Node coordination, technical support, executive administration
C – Energy & Cooling Campus	Reliable and sustainable infrastructure	Substation, UPS, backup power, cooling plant, thermal storage, heat recovery
D – Maintenance & Logistics	Infrastructure support	Technical workshops, spare parts, equipment staging, fleet support
E – Security Perimeter	Critical infrastructure protection	Controlled entrances, security airlocks, access control, CCTV, perimeter protection
F – Innovation & Visitor Center	Ecosystem engagement	Demonstration center, municipal engagement, research collaboration, education and community outreach

9. Site Layout – Operational Interpretation

The uploaded document's cover/site visualization shows a campus organized around a central AI Infrastructure Core, with dedicated operations, maintenance/storage, electrical substation, technical cooling, parking, security/access, employee housing and visitor functions. The visualization also presents solar generation, green landscaping and distributed connectivity as elements of the intended campus concept.

The source image labels the principal elements as: 1 AI Infrastructure Core; 2 Administration & Operations Building; 3 Maintenance & Storage Building; 4 Dedicated 20kV/110kV Substation; 5 Technical Cooling Plant; 6 Parking Area; 7 perimeter/road infrastructure; 8 Main Entrance; 9 Service Vehicle Entrance; 10 Employee Housing Building; 11 Children's Playground; 12 Guest Center.

These visual elements should be treated as masterplan concepts, not detailed construction drawings. Site engineering, zoning, planning permission, fire protection, environmental assessment, traffic access and utility studies remain necessary.

10. Energy and Cooling Architecture

10.1 Electrical Supply

- Dedicated electrical infrastructure is planned as a core site function.
- The source visualization identifies a dedicated 20 kV / 110 kV substation.
- UPS systems and backup power are included in the Energy & Cooling Campus.
- Final grid connection and redundancy architecture require utility and engineering validation.

10.2 Cooling

- Direct-to-Chip liquid cooling is the target technology for high-density AI workloads.
- Dedicated cooling plant infrastructure is planned.
- Thermal storage and heat recovery are included in the source concept.
- Future municipal heat integration is identified as a potential application.

10.3 Sustainability

Target PUE ≤ 1.20

- Heat recovery and potential municipal heat integration
- Energy-efficient cooling
- Potential renewable-energy integration
- Long-life infrastructure design
- Measurement of actual energy and thermal performance

11. Connectivity and Network Architecture

The source specifies multi-carrier fiber connectivity. In the updated masterplan, this is essential not only for conventional data-center operations but also for DSAINT federation workloads.

Network function	Requirement
External connectivity	Multi-carrier, resilient connectivity.
Federation connectivity	Secure connectivity to participating DSAINT nodes.
Management network	Dedicated protected management plane.
Workload network	High-performance network for AI services and distributed workloads.
Security monitoring	Traffic visibility and controlled security boundaries.
Redundancy	Avoid single-carrier or single-path dependency.

12. Security Framework

The uploaded Masterplan identifies BSI IT-Grundschutz, dedicated secure networks, Zero Trust principles, encrypted communications, strict identity and access management, GDPR compliance and state/federal regulations.

- Physical perimeter protection
- Controlled visitor and service access
- Identity and access management
- Network segmentation
- Encrypted communications
- Workload isolation
- Security monitoring and incident response
- Audit logging
- Secure node onboarding
- Security classification for future federation participants

Node 01 should serve as the highest-trust infrastructure tier in the DSAINT federation.

13. Operations Model

Operational function	Primary responsibility
NOC	Infrastructure, network, compute and service monitoring.
SOC	Security monitoring, incident detection and response.
Federation Operations	Node onboarding, workload orchestration, health and

	policy management.
Technical Support	Infrastructure and participant support.
Maintenance	Preventive and corrective maintenance.
Logistics	Spare parts, equipment staging and fleet support.
Administration	Contracts, procurement, compliance and coordination.
Visitor & Innovation	Municipal, research, enterprise and education engagement.

14. Community Compute Federation Integration

The uploaded Masterplan assigns Node 01 responsibility for Citizen Node onboarding, workload orchestration, performance monitoring, contribution accounting, compensation calculation and federation security management.

The updated strategy introduces a staged implementation path. The physical Node 01 should not immediately assume responsibility for thousands of Citizen AI Nodes. Instead, the federation should first be validated through a controlled software-defined prototype and a limited number of trusted nodes.

1. Prototype node identity and registration.
2. Test controlled workload scheduling.
3. Validate monitoring and contribution measurement.
4. Introduce trusted external participants.
5. Test security and isolation.
6. Validate accounting and reporting.
7. Only then evaluate broader Citizen Node participation.

15. Economic and Regional Impact

The source document identifies infrastructure investment, skilled employment, research collaboration, AI ecosystem growth, municipal innovation and community participation as expected economic effects.

- Creation of specialized infrastructure and operations employment
- Research and development opportunities
- Local and regional supplier participation
- Municipal innovation partnerships
- Potential waste-heat utilization
- AI ecosystem attraction and retention
- Long-term participation in distributed AI infrastructure

A detailed economic-impact model, CAPEX/OPEX model and financing structure should be developed separately and should not be inferred from the masterplan alone.

16. Development and Investment Gates

Gate	Decision evidence
Gate 0 – Concept	Strategic need, site concept and partner interest.
Gate 1 – Technical feasibility	Orchestration/federation prototype and initial benchmarks.
Gate 2 – Site feasibility	Land, grid, fiber, water/cooling, planning and environmental studies.

Gate 3 – Business feasibility	Demand, customers, operating model, CAPEX/OPEX and financing.
Gate 4 – Detailed engineering	Electrical, cooling, network, security and building design.
Gate 5 – Construction readiness	Permits, procurement, financing and implementation partners.
Gate 6 – Commissioning	Integrated testing of power, cooling, network, security and compute.
Gate 7 – Operational validation	AI workload, federation, sustainability and reliability KPIs.

The 2027 commissioning date from the source remains a planning target, not a guaranteed delivery date.

17. Updated Development Roadmap

Period	Masterplan milestone	Updated interpretation
2026	Planning, permitting and project preparation	Feasibility, partnerships, technical prototype and site studies.
2027	Commissioning target	Only if technical, regulatory, financing and construction gates are satisfied.
2028	Citizen Node Program / Federation	Controlled pilot based on validated orchestration and security.
2030	Three German Core Nodes	Expansion after evidence from Node 01.
2032	National AI Ecosystem	Broader research, enterprise and community participation.
2034	Twelve German Core Nodes	Long-term national network target.
2036	European Federation	Long-term cross-border expansion target.

The dates are retained from the uploaded source, but the updated masterplan treats them as strategic targets rather than fixed commitments.

18. Key Performance Indicators

Domain	Indicative KPI
Infrastructure	IT capacity delivered vs. design
Energy	Actual PUE and thermal efficiency
Cooling	Cooling availability and thermal performance
Reliability	Availability and redundancy performance
Network	Connectivity availability and latency
AI compute	GPU utilization and workload throughput
Orchestration	Scheduling success rate and placement efficiency
Federation	Successful node onboarding and workload completion
Security	Security incidents, response times and policy compliance
Operations	Mean time to detect / recover from incidents

Sustainability	Recovered heat and renewable-energy contribution where applicable
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These KPIs are proposed management measures; the uploaded Masterplan does not specify numerical targets beyond the 2 MW/6 MW capacity, PUE ≤ 1.20 and stated infrastructure characteristics.

19. Principal Risks and Mitigations

Risk	Mitigation / control
High capital requirement	Stage investment behind technical and business feasibility gates.
Grid connection constraints	Begin utility studies early; evaluate redundancy and expansion path.
Cooling complexity	Validate liquid-cooling architecture through engineering and pilot testing.
Technology changes	Use modular infrastructure and standards-based interfaces.
Cybersecurity	Security-by-design, Zero Trust principles and controlled federation.
Underutilization	Secure anchor demand and validate workload use cases before expansion.
Federation complexity	Start with trusted nodes and progressively increase participation.
Regulatory/permitting delays	Dedicated permitting and compliance workstream.
Construction schedule	Use stage gates and experienced EPC/engineering partners.

20. Required Partner Ecosystem

- Municipality / regional authorities
- Grid and energy partners
- Data-center engineering and construction partners
- Liquid-cooling specialists
- Network and connectivity providers
- AI/GPU infrastructure operators
- Universities and research institutions
- Cybersecurity and compliance specialists
- AI software and orchestration specialists
- Industrial anchor customers
- Potential heat-recovery partners
- Future federation participants

DSAIN'T's current strategy is explicitly team-oriented. Node 01 should be developed with experienced partners rather than as a founder-only engineering effort.

21. Relationship to DSAIN'T Federation Prototype 01

The physical masterplan and the software prototype are complementary.

Software prototype	Node 01 physical implementation
Virtual / small physical nodes	High-density production AI infrastructure
Scheduling algorithms	Production workload placement
Identity and trust model	Physical access + cyber identity
Monitoring	NOC/SOC and infrastructure telemetry
Failure handling	Redundant power, cooling, network and compute
Contribution accounting	Federation operations and future compensation
Policy enforcement	Production security and governance

Prototype first → validate → engineer → construct → integrate → operate

22. Masterplan Deliverables

- Site feasibility and constraints study
- Conceptual campus layout
- Electrical architecture
- Cooling and thermal architecture
- Network architecture
- Security architecture
- Building and operational zoning
- AI infrastructure design
- DSAINTE orchestration integration specification
- Federation operations specification
- Environmental and sustainability concept
- CAPEX/OPEX financial model
- Permitting and compliance plan
- Procurement and implementation plan
- Commissioning and operational readiness plan

23. Long-Term Role of Demonstrator Node 01

The uploaded Masterplan describes Node 01 as more than a data center: it is intended to be the first building block of Germany's sovereign AI federation. That principle remains the central strategic role of the project.

The updated Masterplan makes the path to that role more rigorous. Node 01 should become the reference environment where DSAINTE's distributed infrastructure, workload orchestration, trusted federation, security, sustainability and operational concepts are demonstrated under real conditions.

If validated, the resulting architecture can provide a repeatable blueprint for additional German Core Infrastructure Nodes and, ultimately, a European sovereign AI federation.

BUILD THE REFERENCE NODE. VALIDATE THE FEDERATION. CREATE THE BLUEPRINT FOR SCALE.

24. Contact and Document Status

Item	Information
Project	DSAINTE – Demonstrator Node 01
Program	DSAINTE Sovereign AI Infrastructure Ecosystem
Location	Bedburg-Hau, Germany
Founder	Ernest Repko
Email	info@dsaint.eu
Website	www.dsaint.eu
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Concept Masterplan – not an offer of securities or investment solicitation.